



AI-Powered Indian Health-Tech Platform

Complete Project Architecture

Technical Documentation & Structure Analysis

6

Artifacts

100+

API Endpoints

40+

DB Tables

5

Languages

Platform Components

Mobile App

Expo React Native

Landing Page

React + Vite

Admin Panel

React + Vite

Business Portal

React + Vite

API Server

Express.js + Drizzle ORM

Hosted on Render (Mumbai region)

Database

Supabase PostgreSQL (Mumbai)

40+ tables, Drizzle ORM migrations

Technology Stack

TypeScript

pnpm Monorepo

NVIDIA AI

Razorpay

Resend

Firebase

Deployment: Render (API) • Vercel (Admin + Landing) • Expo EAS (Mobile)

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Aorane Platform — Technical Architecture

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1. Project Overview

Mission, Architecture & Platform Description

What is Aorane?

Aorane is an AI-powered Indian health-tech platform that provides personalized health management for individual users and enterprise wellness solutions for organizations (corporates, hospitals, gyms). The platform combines real-time health tracking, AI-driven food analysis, stress management, blood bank coordination, emergency SOS, and B2B corporate wellness — all tailored for the Indian market with multilingual support (Hindi/English), Indian dietary patterns, and local health context.

Platform Pillars

📱 Mobile Health App

Expo React Native app for individual users — food tracking, medicine reminders, stress monitoring, AI scan, health reports, scorecard

🏢 B2B Business Portal

Web dashboard for organizations — employee enrollment, aggregate health analytics, wellness programs, billing management

⚙️ Admin Panel

Internal ops tool — user management, revenue tracking, feature flags, AI config, ads, promo codes, support tickets, audit logs

🌐 Landing Website

Public-facing marketing site — product features, pricing, business solutions, blogs, contact, careers

🖥️ API Server

Central Express.js backend — JWT auth, 100+ REST endpoints, Drizzle ORM, Redis caching, AI integrations, webhooks

🗄️ Supabase Database

PostgreSQL on Supabase Mumbai — 40+ tables, migration-based schema, full-text search, real-time capabilities

Target Users

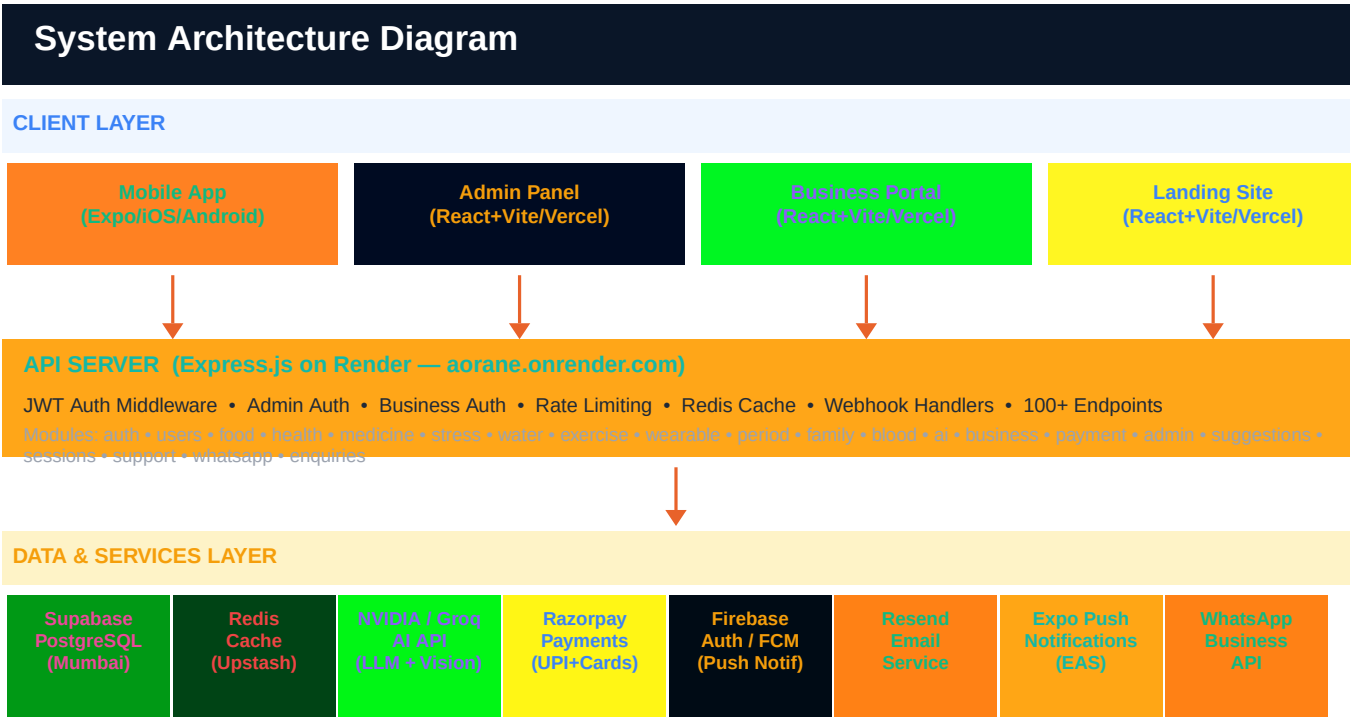
- Individual users seeking personal health tracking
- Corporate employees on company wellness programs
- Hospital patients & health-conscious professionals
- Gym members & fitness enthusiasts
- Families using shared health plans
- NGOs & schools running health programs
- Insurance companies monitoring member health
- Blood donors & emergency responders

Key Differentiators

- AI-powered food scan from photo/text/voice
- Unique 12-digit Aorane ID for every user
- Indian food database with local nutritional data
- Hindi + English AI response generation
- Corporate B2B enrollment with seat-based billing
- Blood bank network with emergency SOS alerts
- Google Fit / wearable device integration
- WhatsApp bot for health check-ins

2. System Architecture & 3. Technology Stack

Component diagram, connections, and full tech stack



3. Technology Stack		
Layer	Technology	Purpose
Mobile App	Expo SDK 52, React Native, TypeScript	iOS + Android + Web
Web Apps	React 18, Vite 7, TypeScript, TailwindCSS	Admin, Business, Landing
API Server	Node.js 20, Express.js, TypeScript	REST API backend
ORM	Drizzle ORM + drizzle-kit	DB schema & migrations
Database	PostgreSQL 15 via Supabase	Primary data store
Caching	Redis via Upstash	OTP, sessions, rate limits
AI / LLM	NVIDIA NIM (LLaMA 3.3 70B)	Food scan, health tips, chat
Auth	JWT (access + refresh), bcrypt, Firebase	Token-based multi-platform
Payments	Razorpay Orders + Subscriptions	UPI, cards, autopay
Email	Resend API	OTP, invoices, onboarding
Push	Expo Push + Firebase FCM	Mobile notifications
Wearable	Google Fit OAuth2	Steps, calories, heart rate
Build	pnpm workspaces, esbuild	Monorepo + fast builds
Deploy API	Render (ap-south-1)	Auto-deploy on git push
Deploy Web	Vercel	CDN, serverless, preview URLs
Monorepo	pnpm workspace + TypeScript refs	Shared code, type safety

4. Deployment Infrastructure

Hosting, CI/CD, domains, and environment configuration

Deployment Overview			
Service	Platform	Domain / Region	Auto-Deploy
API Server	Render	aorane.onrender.com (Mumbai)	' Git push to main
Admin Panel	Vercel	admin.aorane.in	' Git push to main
Landing Page	Vercel	aorane.in	' Git push to main
Business Portal	Vercel	business.aorane.in	' Git push to main
Mobile App	Expo EAS	App Store + Play Store	Manual EAS build
Database	Supabase	ap-south-1 (Mumbai)	Always-on managed
Cache/Redis	Upstash	Global edge CDN	Always-on managed
Email	Resend	Global CDN	API-based
Domain	aorane.in	Cloudflare DNS	Managed separately

Environment Variables & Secrets		
Variable	Used In	Purpose
SUPABASE_DATABASE_URL	API Server	PostgreSQL connection string
SUPABASE_URL	API Server	Supabase REST/realtime endpoint
SUPABASE_ANON_KEY	API Server	Public Supabase API key
SESSION_SECRET	API Server	JWT signing secret
RAZORPAY_KEY_ID	API Server	Razorpay public key
RAZORPAY_KEY_SECRET	API Server	Razorpay secret key
RAZORPAY_WEBHOOK_SECRET	API Server	Webhook signature verification
NVIDIA_API_KEY	API Server	NVIDIA NIM LLM API access
RESEND_API_KEY	API Server	Email delivery via Resend
GOOGLE_FIT_CLIENT_ID	API Server	Google Fit OAuth2 client
GOOGLE_FIT_CLIENT_SECRET	API Server	Google Fit OAuth2 secret
EXPO_TOKEN	EAS Build	Expo build & submission token

Monorepo Structure	
artifacts/aorane-mobile/	– Expo React Native mobile app
artifacts/admin-panel/	– React+Vite admin dashboard (Vercel)
artifacts/business-portal/	– React+Vite B2B portal (Vercel)
artifacts/aorane-landing/	– React+Vite marketing site (Vercel)
artifacts/api-server/	– Express.js backend (Render)
lib/db/	– Drizzle ORM schema + migrations (shared)
lib/api-spec/	– OpenAPI specification
lib/api-zod/	– Generated Zod validation schemas
lib/api-client-react/	– Generated React Query hooks
scripts/	– Utility scripts (seed, PDF gen)
pnpm-workspace.yaml	– Workspace packages + catalog pins
tsconfig.base.json	– Shared TypeScript strict config

5. API Server — Complete Route Reference

100+ REST endpoints across 22 modules

5.1 Authentication (auth.ts)

Mobile login via OTP (SMS/WhatsApp), Google OAuth, Firebase, PIN-based re-auth, refresh tokens

POST	/auth/send-otp	Send OTP to phone number (Firebase SMS or direct Twilio)
POST	/auth/send-otp-whatsapp	Send OTP via WhatsApp
POST	/auth/verify-otp	Verify OTP !' returns access + refresh JWT
POST	/auth/send-email-otp	Send OTP to email address
POST	/auth/verify-email-otp	Verify email OTP
POST	/auth/google	Google OAuth login (ID token exchange)
POST	/auth/firebase-login	Firebase custom token login
POST	/auth/pin/set	Set 4/6-digit security PIN
POST	/auth/pin/login	Login with PIN (returns new JWT)
POST	/auth/refresh	Refresh access token using refresh token
POST	/auth/logout	Invalidate refresh token
GET	/auth/me	Get current authenticated user profile

5.2 User Management (users.ts)

Profiles, preferences, privacy, health goals, medical conditions, activity scores

GET	/users/profile	Get full user profile with stats
PATCH	/users/profile	Update profile (name, DOB, city, language, etc.)
PATCH	/users/onboarding/step	Complete onboarding step (1–5)
GET	/users/preferences	Get app preferences (dark mode, reminders, etc.)
PATCH	/users/preferences	Update preferences
GET	/users/privacy	Get privacy settings
PATCH	/users/privacy	Update privacy settings
POST	/users/health-goals	Set health goals (weight loss, fitness, etc.)
POST	/users/medical-conditions	Add/update medical conditions
GET	/users/scorecard	Get Aorane ID card with QR code
GET	/users/activity-score	Get 7-day activity score breakdown
GET	/users/search	Search users by phone/Aorane ID (limited)
POST	/users/push-token	Register Expo push notification token

5.3 Food & Nutrition (food.ts)

AI food scan, manual logging, history, favorites, weekly nutrition aggregates

POST	/food/scan	AI-powered food scan (photo/text/voice input !' nutrition)
POST	/food/log	Log a food entry manually or from scan
GET	/food/logs	Get food logs by date range
DELETE	/food/log/:id	Delete a food log entry
GET	/food/summary/:date	Get daily nutrition summary for date
GET	/food/weekly-nutrition	Get 7-day nutrition totals + micronutrients
GET	/food/search	Search food items database
GET	/food/history-search	Search user's food history
GET	/food/favorites	Get user's favorite foods
GET	/food/db-stats	Stats on food database size
POST	/food/weather-suggestions	AI suggestions based on weather + location

5.4 Health Tracking (health.ts, stress.ts, water logs)

Exercise, water, stress logs, daily & weekly health scores, predictions

POST	/health/exercise	Log exercise session (type, duration, calories)
GET	/health/exercise	Get exercise history
POST	/health/exercise/calculate	Calculate calories for exercise
POST	/health/water	Log water intake (ml)
GET	/health/water/:date	Get water intake for date
GET	/health/score/:date	Get daily health score
POST	/health/score/:date/compute	Compute/refresh health score

GET	/health/scores/history	Health score history (30 days)
GET	/health/weekly-activity	Weekly activity summary
GET	/stress/logs	Get stress logs
POST	/stress/log	Log stress (PPG/mood/5-pillar)
GET	/stress/today	Today's stress summary
GET	/stress/weekly	Weekly stress trend
GET	/stress/insight	AI-generated stress insight

5.5 Medicine & Wearable (medicine.ts, wearable.ts)

Medicine schedules, reminders, logs, Google Fit OAuth, wearable data sync

POST	/medicine/schedule	Create medicine schedule
GET	/medicine/schedules	Get all medicine schedules
PATCH	/medicine/schedule/:id	Update schedule (name, timing, dose)
GET	/medicine/today	Today's medicine reminders
POST	/medicine/log	Log medicine taken
GET	/medicine/logs	Get medicine logs
GET	/wearable/providers	List supported wearable providers
GET	/wearable/oauth/google-fit/url	Get Google Fit OAuth URL
GET	/wearable/oauth/google-fit/callback	OAuth callback handler
GET	/wearable/connections	Get connected wearables
POST	/wearable/sync/:provider	Sync data from wearable
POST	/wearable/data/manual	Manually add wearable data
GET	/wearable/data	Get synced wearable data
DELETE	/wearable/connections/:provider	Disconnect wearable

5.6 AI & Intelligence (ai.ts, intelligence.ts, suggestions.ts)

NVIDIA LLM-powered: diet plans, meal swaps, health chat, risk assessments, daily suggestions

POST	/ai/smart-scan	AI food analysis from image/text/voice
POST	/ai/diet-plan	Generate personalized AI diet plan
POST	/ai/meal-swap	Suggest healthier meal alternatives
POST	/ai/health-tip	Generate contextual health tip
GET	/health/intelligence/predict	AI health risk prediction (cached)
POST	/health/intelligence/predict/refresh	Refresh AI health prediction
GET	/health/intelligence/diet-chart	Weekly AI-generated diet chart
POST	/health/intelligence/diet-chart/refresh	Refresh diet chart
GET	/health/intelligence/exercise/met	MET values for exercise types
POST	/health/intelligence/exercise/calories	Calculate calories from MET
GET	/suggestions/daily	AI daily suggestions (cached 24h)
POST	/suggestions/refresh	Force refresh daily suggestions
GET	/notifications/settings	Notification preferences
PUT	/notifications/settings	Update notification preferences

5.7 Blood Bank & Emergency (blood.ts, accident-emergency.ts)

Blood donor registration, emergency requests, donor matching, accident SOS, push alerts

POST	/blood/donor/register	Register as blood donor
POST	/blood/donor/verify-otp	Verify donor OTP
GET	/blood/request	Search blood requests nearby
POST	/blood/request	Create emergency blood request
POST	/blood/request/verify-otp	OTP verification for blood request
POST	/blood/request/:id/respond	Donor responds to request
POST	/blood/request/:id/flag	Flag/resolve blood request
POST	/blood/emergency/direct	Direct blood emergency SOS
POST	/blood/donation	Log blood donation
GET	/blood/donations	Get donation history
POST	/emergency/accident/sos	Accident SOS trigger
GET	/emergency/accident/history	Accident alert history
PATCH	/emergency/accident/:id/cancel	Cancel accident alert

PATCH	/emergency/accident/:id/resolve	Resolve accident alert
GET	/emergency/contacts	Get emergency contacts
POST	/emergency/contacts	Add emergency contact
DELETE	/emergency/contacts/:id	Remove emergency contact

5.8 Business / B2B (business.ts)

Organization registration, login, member enrollment, analytics, announcements, billing

POST	/business/register	Register new organization
POST	/business/login	Organization admin login
POST	/business/login/send-email-otp	Send email OTP for login
POST	/business/login/verify-otp	Verify email OTP !' JWT
POST	/business/forgot-password	Send password reset OTP
POST	/business/forgot-password/verify	Verify reset OTP + new password
GET	/business/me	Get organization details
GET	/business/overview	Overview stats (members, seats, usage)
GET	/business/analytics	Aggregate health analytics for org
GET	/business/health-analytics	Detailed health trend analytics
GET	/business/members	List all org members
GET	/business/members/search	Search members by name/ID
GET	/business/members/suspended	List suspended members
GET	/business/members/:userId/detail	Member health detail view
GET	/business/members/:userId/stress	Member stress trends
POST	/business/enroll	Enroll user via enrollment code
GET	/business/enrollment-codes	Get enrollment codes
POST	/business/enrollment-codes	Generate new enrollment code
GET	/business/announcements	Get org announcements
POST	/business/announcements	Post announcement to members
PATCH	/business/settings	Update org settings
PATCH	/business/admin/password	Change admin password
GET	/business/billing/plans	Get available plans
GET	/business/billing/seat-plans	Get seat-based pricing
GET	/business/billing/subscription	Get current subscription
POST	/business/billing/order	Create Razorpay order
POST	/business/billing/verify	Verify payment
POST	/business/billing/subscription/create	Create subscription
POST	/business/billing/subscription/verify	Verify subscription payment
POST	/business/billing/seat-order	Create seat upgrade order
POST	/business/billing/seat-verify	Verify seat payment
GET	/business/billing/invoices	Get billing invoices
DELETE	/business/billing/subscription/cancel	Cancel subscription

5.9 Payments & Revenue (payment.ts, webhook.ts)

Razorpay orders, subscriptions, promo codes, checkout pages, webhook verification

POST	/payment/order	Create one-time Razorpay order
POST	/payment/verify	Verify one-time payment signature
POST	/payment/promo/validate	Validate promo code
GET	/payment/subscription	Get current subscription status
POST	/payment/subscription/create	Create autopay subscription
POST	/payment/subscription/verify	Verify first subscription payment
DELETE	/payment/subscription/cancel	Cancel subscription
GET	/payment/checkout/:orderId	Server-rendered Razorpay checkout (mobile)
GET	/payment/subscription-checkout/:id	Server-rendered subscription checkout
GET	/payment/rzp-callback	Razorpay payment callback (GET)
POST	/payment/rzp-callback	Razorpay payment callback (POST)
POST	/payment/subscription-rzp-callback	Subscription callback (POST)
GET	/payment/order-status	Poll payment order status

GET	/payment/razorpay-test	Test Razorpay connectivity
POST	/webhooks/razorpay	Razorpay webhook (HMAC verified)

5.10 Admin Panel Routes (admin.ts, plans.ts, ads.ts)

User management, org management, revenue, feature flags, food DB, AI config, promo codes

POST	/admin/login	Admin login (rate-limited, bcrypt)
GET	/admin/me	Get admin profile
PATCH	/admin/me	Update admin profile
POST	/admin/change-password	Change admin password
GET	/admin/overview	Platform overview stats
GET	/admin/analytics	Detailed analytics (DAU, revenue, etc.)
GET	/admin/users	Paginated user list with profiles
PATCH	/admin/users/:id	Update user (plan, ban, activate)
GET	/admin/users/search	Search users (UUID/phone/email/name)
GET	/admin/organizations	List all organizations
PUT	/admin/organizations/:id	Edit organization details
DELETE	/admin/organizations/:id	Delete organization
PATCH	/admin/organizations/:id/toggle-active	Activate/deactivate org
GET	/admin/subscriptions	List all subscriptions
PATCH	/admin/subscriptions/:id/cancel	Cancel user subscription
GET	/admin/revenue	Revenue breakdown (daily/monthly)
GET	/admin/platform-costs	Platform cost tracking
GET	/admin/feature-flags	List all feature flags
POST	/admin/feature-flags	Create feature flag
PATCH	/admin/feature-flags/:key	Toggle/update feature flag
GET	/admin/food-items	Manage food items database
GET	/admin/food-cache	AI food scan cache entries
POST	/admin/food-cache/:id/promote	Promote cache entry to food DB
POST	/admin/food-cache/:id/reject	Reject cache entry
GET	/admin/food-cache/stats	Food cache statistics
GET	/admin/food-cache/export	Export food cache as CSV
GET	/admin/promo-codes	List promo codes
GET	/admin/announcements	List announcements
POST	/admin/announcements	Create platform announcement
GET	/admin/support-tickets	List support tickets
PATCH	/admin/support-tickets/:id	Update ticket status
GET	/admin/blood-requests	View blood emergency requests
PATCH	/admin/blood-requests/:id	Update blood request status
GET	/admin/audit-logs	Admin action audit trail
GET	/admin/languages	List supported languages
GET	/admin/ai-config	Get AI model configuration
PUT	/admin/ai-config	Update AI model config
GET	/admin/settings/company	Get company settings
PUT	/admin/settings/company	Update company settings
GET	/admin/org-invoices	List B2B org invoices
GET	/plans	Get plan pricing (public)
PUT	/admin/plan-pricing/:planKey	Update plan pricing
POST	/admin/plan-pricing/reset	Reset plan pricing to defaults
GET	/admin/whatsapp/config	WhatsApp bot config
PUT	/admin/whatsapp/config	Update WhatsApp bot config
GET	/admin/whatsapp/stats	WhatsApp message stats
GET	/admin/whatsapp/templates	WhatsApp message templates

5.11 Misc: Period, Family, Sessions, Support, WhatsApp, Enquiries

Women's health, family groups, app sessions (DAU), help tickets, WhatsApp bot

POST	/period/log	Log menstrual cycle data
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GET	/period/logs	Get period history
PATCH	/period/log/:id	Update period log
GET	/family/group	Get family group details
POST	/family/group	Create family group
POST	/family/invite	Invite member to family
POST	/family/join	Join family with invite code
GET	/family/members	List family members
POST	/sessions/start	Start app session
POST	/sessions/heartbeat	Session heartbeat (keep alive)
POST	/sessions/end	End app session
GET	/sessions/me	Get user's session history
GET	/sessions/dau	Daily active users count (admin)
POST	/support/ticket	Create support ticket
GET	/support/tickets/mine	Get user's own tickets
POST	/whatsapp/subscribe	Subscribe to WhatsApp health updates
DELETE	/whatsapp/unsubscribe	Unsubscribe from WhatsApp
GET	/whatsapp/status	Get WhatsApp subscription status
GET	/whatsapp/webhook	WhatsApp webhook verification
POST	/whatsapp/webhook	Receive WhatsApp messages
GET	/whatsapp/subscription	Get subscription details
PUT	/whatsapp/preferences	Update message preferences
GET	/whatsapp/message-history	Get message history

6. Database Schema

PostgreSQL on Supabase — 40+ tables across 8 schema files using Drizzle ORM

6.1 Users & Profiles (users.ts)

Table Name	Key Columns / Purpose
users	id(uuid), phone, email, plan(free/pro/max/family), isActive, isBanned, createdAt
user_auth_providers	userId, provider(mobile/google/facebook/x), providerUid, email
user_profiles	userId, aoraneId(12-digit unique), fullName, DOB, gender, BMI, city, state, bloodGroup
user_medical_conditions	userId, condition, severity, diagnosedAt, notes
user_health_goals	userId, goalType(weight/fitness/etc), targetValue, currentValue, deadline
user_preferences	userId, languageCode, darkMode, waterGoalGlasses, calorieGoal, notification prefs
user_privacy_settings	userId, profileVisibility, shareHealthData, allowContact settings
otp_store	phone, hashedOtp, expiresAt — temporary OTP store with Redis backup

6.2 Health Tracking (health-tracking.ts)

Table Name	Key Columns / Purpose
exercise_logs	userId, exerciseType, durationMins, caloriesBurned(string), intensity, loggedAt
water_logs	userId, mlAmount, loggedAt, inputMethod(manual/wearable)
medicine_schedules	userId, medicineName, dosage, frequency, reminderTimes[], isActive
medicine_logs	userId, scheduleId, takenAt, skipped, snoozeCount
stress_logs	userId, stressType(ppg/mood/5pillar), score, mood, pillars{}, loggedAt
period_logs	userId, startDate, endDate, cycleLength, symptoms[], notes
medical_reports	userId, reportType, fileUrl, extractedData, uploadedAt
daily_health_scores	userId, date, score(0-100), breakdown{food,water,exercise,medicine,sleep}
daily_suggestions	userId, date, suggestions[], generatedBy(AI), language
health_predictions	userId, riskType, riskLevel, confidence, factors[], generatedAt
weekly_diet_charts	userId, weekStart, meals{}, macros{}, generatedAt

6.3 Food & Nutrition (health-food.ts)

Table Name	Key Columns / Purpose
food_items	id, nameEn, nameHi, calories, protein, carbs, fat, calcium, vitaminC, vitaminB12, iron, fiber, mealType, isVerified
food_scan_cache	query, imageHash, result{nutrients}, source, hitCount, lastHit — AI scan cache
food_logs	userId, foodItemId, foodNameEn, calories, protein, fat, carbs, vitaminC, calcium, vitaminB12, iron, servingSize, mealType, loggedAt, inputMethod

6.4 Business & Organizations (business.ts)

Table Name	Key Columns / Purpose
organizations	id, name, orgType(corporate/hospital/gym/etc), contactEmail, city, state, totalSeats, usedSeats, plan, isActive
org_admins	id, orgId, email, hashedPassword, name, role — portal login credentials
org_members	userId, orgId, role, enrolledAt, isActive — links users to orgs
enrollment_codes	orgId, code, maxUses, usedCount, expiresAt, isActive
org_payments	orgId, amount, planType, seats, razorpayOrderId, status
org_announcements	orgId, title, body, targetRoles, sentAt
insurance_api_keys	orgId, provider, apiKey, isActive — for insurance integrations

6.5 Revenue & Payments (revenue.ts)

Table Name	Key Columns / Purpose
subscriptions	userId, plan, status, amount, expiresAt, razorpaySubscriptionId, autoRenew, promoUsed
payments	userId, amount, plan, status, razorpayOrderId, razorpayPaymentId, createdAt
promo_codes	code, discountPercent, maxUses, usedCount, expiresAt, isActive, createdBy
plan_pricing	planKey(free/pro/max/family), monthlyPrice, yearlyPrice, features[]
referrals	referrerId, refereeId, code, status, rewardGiven, createdAt

6.6 Platform & Admin (platform.ts)

Table Name	Key Columns / Purpose
push_tokens	userId, token, platform(ios/android), isActive, lastSeenAt
notifications	userId, type, title, body, data{}, isRead, sentAt
announcements	title, body, targetPlan, targetOrgType, scheduledAt, sentAt — platform-wide
feature_flags	key, isEnabled, rolloutPercent, description, updatedAt
ad_campaigns	title, type(google/direct), imageUrl, targetUrl, budget, status, impressions, clicks
admin_users	email, hashedPassword, name, role, lastLoginAt
admin_audit_logs	adminId, action, targetType, targetId, changes[], ip, createdAt
ai_config	provider, model, temperature, maxTokens, systemPrompt, isActive
company_settings	key, value — platform-wide configuration KV store
enquiries	name, email, phone, subject, message, status, createdAt
app_sessions	userId, sessionId, startedAt, endedAt, duration, platform, appVersion

6.7 Community & Emergency (community.ts)

Table Name	Key Columns / Purpose
family_groups	id, name, ownerId, inviteCode, maxMembers, createdAt
family_members	groupId, userId, role(owner/member), joinedAt
blood_donors	userId, bloodGroup, city, isAvailable, lastDonatedAt, donationCount, otpVerified
blood_emergency_requests	requesterId, bloodGroupNeeded, unitsNeeded, hospitalName, city, status, expiresAt
blood_emergency_responses	requestId, donorId, response(can_help/later/unavailable)
blood_donations	donorId, requestId, donatedAt, hospitalName, units
accident_emergency_logs	userId, lat, lng, status, contacts[], resolvedAt
emergency_contacts	userId, name, phone, relationship, notifyOnSOS

6.8 Wearable, Languages & WhatsApp (wearable.ts + whatsapp.ts)

Table Name	Key Columns / Purpose
wearable_connections	userId, provider(google-fit), accessToken, refreshToken, lastSyncAt
wearable_data	userId, provider, dataType(steps/hr/calories), value, recordedAt
offline_queue	userId, action, payload, retryCount, createdAt — offline sync queue
languages	code, name, nativeName, isActive — supported UI/AI languages
translations	languageCode, key, value — i18n translation strings
whatsapp_bot_config	orgId, token, phoneNumberId, webhookSecret, isActive
whatsapp_subscriptions	userId, phone, status, preferences{}, subscribedAt
whatsapp_message_logs	userId, direction, type, content, timestamp, status
whatsapp_templates	name, language, components[], status — approved message templates

7. Mobile Application (Expo React Native)

iOS + Android + Web • File-based routing (Expo Router) • TypeScript

7.1 App Navigation Structure

The mobile app uses Expo Router (file-based routing) with nested layouts. Authentication state controls routing between onboarding, auth, and main app flows.

ONBOARDING FLOW (app/(onboarding)/)

File	Description
index.tsx	Welcome screen — Get started CTA, language selection
physical.tsx	Physical info — age, gender, height, weight
lifestyle.tsx	Lifestyle — activity level, sleep hours, food preference
health.tsx	Health history — existing conditions, medications
goals.tsx	Health goals — weight target, fitness objectives
permissions.tsx	Permissions — notifications, camera, location

AUTHENTICATION (app/(auth)/)

File	Description
index.tsx	Login screen — phone/email input with country code
verify-otp.tsx	OTP verification (SMS/WhatsApp/Email)
setup-pin.tsx	Set 4/6 digit security PIN
verify-pin.tsx	PIN login screen

MAIN TABS (app/(tabs)/)

File	Description
dashboard.tsx	Home dashboard — score, today's summary, quick actions, announcements
food.tsx	Food logging — meal history, add entry, daily nutrition summary
exercise.tsx	Exercise tracking — log workout, weekly activity chart
medicine.tsx	Medicine reminders — schedules, today's doses, history
scan.tsx	AI Scanner — camera/gallery food scan, instant nutrition analysis
profile.tsx	User profile — Aorane ID, stats, settings, subscription

FEATURE SCREENS (app/)

File	Description
health-report.tsx	Health report PDF — weekly/monthly, micronutrients, export
scorecard.tsx	Aorane ID card with QR code — shareable health identity
intelligence.tsx	AI health insights — risk predictions, diet chart, tips
stress.tsx	Stress management — log, trends, AI insights, breathing exercises
water.tsx	Water intake — log glasses, daily goal progress, reminders
period.tsx	Menstrual cycle tracker — calendar, symptoms, predictions
wearable.tsx	Wearable device — Google Fit connect, data sync, steps/HR
blood.tsx	Blood bank — donor registration, emergency requests, find donors
family.tsx	Family health — create group, invite members, shared plans
medical-emergency.tsx	Emergency SOS — accident alert with GPS, contacts notification
upgrade.tsx	Plan upgrade — Pro/Max/Family comparison and payment flow
enrollment.tsx	Corporate enrollment — enter org code, join company wellness
suggestions.tsx	Daily AI suggestions — personalized health tips & actions
notification-settings.tsx	Notification preferences — toggle each notification type
help.tsx	Help & support — FAQ, create ticket, contact info

edit-profile.tsx	Edit profile — update personal info, photo, language
edit-work-profile.tsx	Work profile — occupation, work city, stress at work

7.2 Mobile App Libraries & Key Dependencies

- expo-router — File-based navigation
- expo-camera — AI food scan camera
- expo-notifications — Push notification handling
- @tanstack/react-query — API data fetching & caching
- expo-secure-store — Secure token/PIN storage
- expo-location — GPS for blood/emergency features
- expo-image-picker — Gallery food image selection
- expo-file-system — PDF download & share
- expo-print — Health report PDF generation
- expo-sharing — Share reports via OS share sheet
- expo-av — Audio for voice food input
- react-native-reanimated — Smooth animations
- @react-native-async-storage — Local data persistence
- dayjs — Date/time formatting (Indian locale)

8. Admin Panel

React + Vite • Deployed on Vercel • admin.aorane.in • Login: admin@aorane.com

8.1 Admin Panel Pages	
Page (Component)	Features & Capabilities
Login.tsx	Admin login with email/password, brute-force protected (10 attempts/15min), JWT session
Dashboard.tsx	Platform KPIs — total users, DAU, revenue, new signups, subscription breakdown, recent activity
Users.tsx	User list with Name/Email/Aorane ID columns, search (UUID/phone/email/name), toggle ban/activate, plan management
Organizations.tsx	B2B org management — list with stats, EditOrgModal (all fields), DeleteConfirmModal, toggle active/inactive
Analytics.tsx	Detailed platform analytics — charts for DAU, signups, food scans, payment trends, feature usage
Revenue.tsx	Revenue dashboard — daily/monthly breakdown, subscription vs one-time, promo code impact, org billing
Subscriptions.tsx	All subscriptions — filter by plan/status, cancel subscription, view payment history
PlanPricing.tsx	Plan price management — edit monthly/yearly pricing for free/pro/max/family plans
Invoices.tsx	Invoice list — user and org invoices, PDF generation, payment status tracking
FeatureFlags.tsx	Feature toggle — enable/disable features per plan or org type, rollout percentage control
FoodItems.tsx	Food database management — CRUD for Indian food items, nutritional data editor
AIFoodDiscovery.tsx	AI scan cache — review auto-discovered foods, promote to DB or reject, export CSV
AIConfig.tsx	AI model configuration — provider, model name, temperature, max tokens, system prompt
PromoCodes.tsx	Promo code management — create/edit/expire codes, discount %, usage tracking
Announcements.tsx	Platform announcements — create targeted announcements by plan or org type
AdsManager.tsx	Ad campaign management — Google/direct ads, budget tracking, impressions/clicks
FeatureFlags.tsx	Feature flag management — per-plan feature toggles with rollout percentages
Languages.tsx	Language management — supported AI/UI languages, enable/disable, add translations
BloodRequests.tsx	Blood emergency monitoring — view active requests, update status, admin resolution
SupportTickets.tsx	Customer support — ticket list, assign, update status, internal notes
Enquiries.tsx	Business enquiries — from landing page contact form, status management
AuditLogs.tsx	Admin action audit trail — who did what, when, IP address, change diff
Branding.tsx	Platform branding — logo, colors, company name for white-label
PlatformCosts.tsx	Cost tracking — API costs, hosting, external services usage
UpcomingFeatures.tsx	Roadmap management — planned features, status, priority
Profile.tsx	Admin profile — update name, email, change password

8.2 Admin Panel Architecture	
● Vite 7 + React 18 + TypeScript	● Auto-logout after 30 min inactivity
● Tailwind CSS for styling	● Role-based access (super-admin / ops)
● React Query for API state	● Audit trail for all destructive actions
● JWT auth stored in localStorage	● Responsive layout with sidebar navigation
● Dark mode toggle (system preference sync)	● Toast notifications for all API operations

9. Business Portal 10. Landing Website

B2B org dashboard • React+Vite • Deployed on Vercel

9. Business Portal (business.aorane.in)	
Page	Features
Landing.tsx	Marketing page for businesses — features, pricing, CTA for registration
Register.tsx	Organization registration — name, type, email, password, contact details
Login.tsx	Business admin login — email + password or OTP, JWT session
Verify.tsx	Email verification for new org registration
AuthRedirect.tsx	Handles redirect after OAuth flows
Dashboard.tsx	Org overview — total members, seat usage, active plans, health score distribution
Members.tsx	Employee list — search, filter by plan/status, individual health overview, suspend
EnrollmentCodes.tsx	Manage enrollment codes — generate, set expiry, usage tracking, share link
Analytics.tsx	Aggregate health analytics — trend charts, BMI distribution, step count averages
Communications.tsx	Org announcements — create targeted messages for all/specific members
Billing.tsx	Billing management — current plan, seat count, payment history, upgrade options
Settings.tsx	Org settings — name, contact, GSTIN, city, industry, password change
not-found.tsx	404 page for invalid routes

B2B Enrollment Flow	
1	Organization registers on business.aorane.in
2	Admin generates enrollment codes (capacity-limited)
3	Codes shared with employees via email/chat
4	Employee enters code in mobile app !' Enrollment screen
5	API validates code !' links user to org !' upgrades plan
6	Employee now has Pro/Max plan funded by their organization
7	Org analytics aggregate employee health data (anonymized)
8	Org pays per-seat billing on Razorpay subscription

10. Landing Website (aorane.in)	
Page	Content
LandingPage.tsx	Hero section, platform stats, key features overview, app store download CTAs
FeaturesPage.tsx	Detailed feature list — AI scan, health tracking, blood bank, B2B, wearable
PricingPage.tsx	Plan comparison table — Free / Pro / Max / Family with feature matrix
BusinessPage.tsx	B2B landing — corporate wellness value prop, case studies, org plan pricing
AboutPage.tsx	About Aorane — mission, team, vision for Indian health-tech
ContactPage.tsx	Contact form (enquiries table), email, office address
PrivacyPage.tsx	Privacy policy — data collection, storage, sharing, deletion rights
TermsPage.tsx	Terms of service — usage rules, subscription terms, cancellation policy
CareersPage.tsx	Jobs page — open positions, team culture, apply CTA
ComingSoonPage.tsx	Coming soon placeholder for unannounced features

11. Security Architecture

Authentication, authorization, encryption, and security hardening

11.1 Authentication System

Layer	Implementation
User Auth (Mobile)	JWT access token (15min) + refresh token (30 days). OTP via SMS/WhatsApp/Email for login
PIN Auth (Mobile)	4/6 digit PIN stored bcrypt-hashed in DB. Quick re-auth without OTP
Google OAuth	Firebase ID token verified server-side, mapped to user account
Firebase Auth	Custom token flow for seamless Google/Firebase login
Business Auth	Email + password (bcrypt) or email OTP for org portal login
Admin Auth	Email + bcrypt password, JWT, rate-limited (10 attempts/15 min)
Token Storage	Expo SecureStore on mobile (hardware keychain). Never localStorage
Token Refresh	Silent refresh on 401 response. Interceptor pattern in API client

11.2 API Security Measures

- All user routes: requireAuth middleware (JWT verify)
- All admin routes: requireAdmin (separate admin JWT)
- All business routes: requireBusinessAuth (org JWT)
- Rate limiting on auth endpoints (Redis-backed)
- Admin login: 10 attempts per 15 min per email
- OTP: time-limited (5 min expiry), single-use, hashed
- SQL injection: Drizzle ORM parameterized queries only
- XSS: safePlanLabel() whitelist + escapeJs() in templates
- CORS: configured per environment (production restricts origins)
- Webhook: Razorpay HMAC-SHA256 signature verification
- Helmet.js security headers on all responses
- Request body size limits to prevent DoS

11.3 Data Security

- Passwords: bcrypt (12 rounds) — never stored plain
- OTPs: SHA-256 hashed in DB + Redis TTL expiry
- Secrets: All in Render/Vercel env vars (never in code)
- DB connection: SSL enabled for Supabase PostgreSQL
- Health data: userId-scoped, no cross-user data leaks
- Push tokens: stored per user, invalidated on logout
- File uploads: Supabase storage with signed URLs
- Audit trail: every admin action logged with IP + changes
- Phone numbers: never logged in plain text in application logs
- Payment: Razorpay handles card data (PCI-DSS compliant)

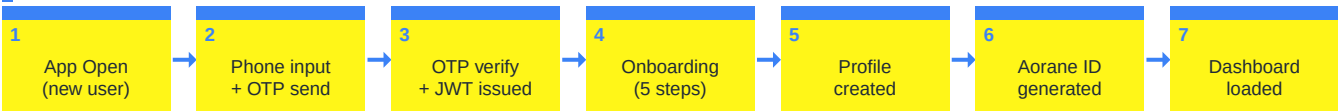
11.4 Infrastructure Security

- HTTPS everywhere — Render, Vercel, Supabase
- TLS 1.2+ enforced on all endpoints
- Supabase Row-Level Security (RLS) policies
- Render private network for DB connections
- No direct DB port exposure — only via app layer
- GitHub Actions CI — no deployment without passing checks
- Dependency audit: 0 critical vulnerabilities
- SAST scan: XSS/SQLi issues fixed in payment + admin
- Secret scan (HoundDog): 0 hardcoded secrets
- google-services.json: Firebase keys restricted by SHA

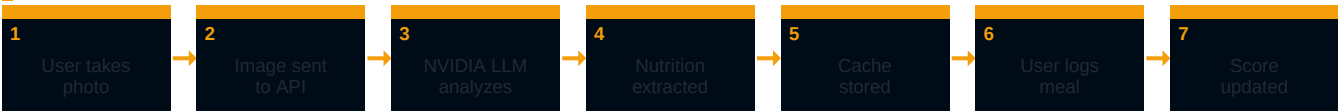
12. Data Flow Diagrams

Key user journeys and data flows through the platform

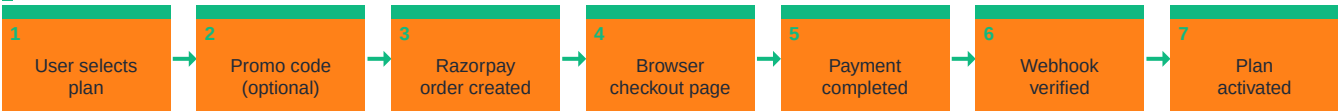
User Registration & Onboarding Flow



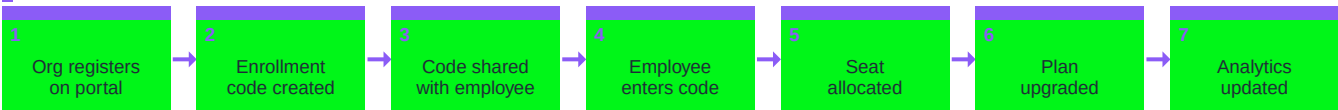
AI Food Scan Flow



Payment & Subscription Flow



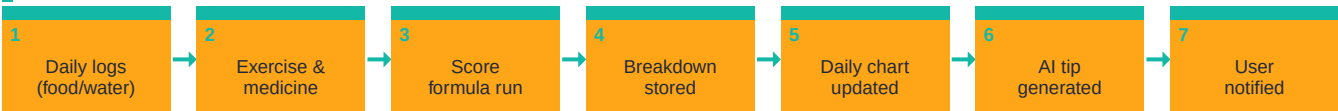
Corporate B2B Enrollment Flow



Blood Emergency Flow



Health Score Computation Flow



Accident Emergency SOS Flow



12.1 Data Architecture Summary

All data is user-scoped at the database level — every table query includes a `userId` `WHERE` clause enforced by middleware. Health data never crosses between users. B2B analytics are aggregated server-side with user PII stripped before sending to org dashboards. Admin-level data access is gated behind a separate auth token and audit-logged.

13. Third-Party Integrations

External services, APIs, and platform integrations

13.1 Integration Summary			
Service	Provider	Purpose	Implementation
AI / LLM	NVIDIA NIM	Food analysis, health tips, diet charts, chat	NVIDIA_API_KEY != callAI() in lib/ai.ts using LLaMA 3.3 70B
Payments	Razorpay	UPI, cards, autopay subscriptions, B2B	RAZORPAY_KEY_ID/SECRET != createOrder(), createSubscription(), webhooks
Email	Resend	OTP emails, invoices, subscription alerts	RESEND_API_KEY != resend.emails.send() for all transactional email
Push Notifications	Expo + Firebase	Mobile push alerts for health reminders	Expo Push API + FCM for Android, APNs for iOS
Auth (Google)	Firebase	Google Sign-In for mobile and web auth	Firebase ID token verification server-side
SMS OTP	Firebase SMS	Phone number verification at login	Firebase Auth phone verification flow
Wearable	Google Fit	Steps, heart rate, calories from exercise	OAuth2 GOOGLE_FIT_CLIENT_ID/SECRET != /wearable/oauth/google_fit
WhatsApp	WhatsApp Business API	Health check-ins, OTP blood sugar	Webhook + message templates via Meta Business API
Database	Supabase	PostgreSQL + storage, realtime	SUPABASE_DATABASE_URL + SUPABASE_ANON_KEY
Cache / Redis	Upstash Redis	OTP storage, rate limiting, session	REDIS_URL via ioredis client in lib/redis.ts
CI/CD API	GitHub	Source control, auto-deploy triggers for production	GitHub integration with Render + Vercel webhooks
Domain / DNS	Cloudflare	DNS, CDN, DDoS protection for	Cloudflare proxied DNS for all subdomains
App Distribution	Expo EAS	Mobile app builds, OTA updates, store	EXPO_TOKEN for EAS build + submit commands

13.2 AI Integration Details (NVIDIA NIM)

The platform uses NVIDIA's NIM API with the LLaMA 3.3 70B Instruct model for all AI features. The AI module (artifacts/api-server/src/lib/ai.ts) provides a unified callAI() function with rate limiting, error handling, and response caching via Redis.

AI Feature	How It Works
Food Scan (smart-scan)	Photo/text/voice != nutrition breakdown (calories, protein, fat, carbs, micronutrients) with Indian food knowledge
Diet Plan Generator	Based on BMI, goals, food preferences, medical conditions != 7-day meal plan with Indian foods
Meal Swap	Suggest healthier alternatives to logged meals with similar calories but better nutrition profile
Health Tips	Contextual daily health tips based on user's health data, Indian seasons, and current metrics
Stress Insights	Analyze stress patterns over 30 days != personalized insight with actionable recommendations
Health Predictions	Risk assessment for lifestyle diseases (diabetes, hypertension) based on BMI, exercise, diet trends
Weekly Diet Chart	Full weekly structured meal plan with recipes, prep time, and nutrient targets
Daily Suggestions	Morning AI-generated health checklist — water goal, step target, meal suggestions, reminders

13.3 Razorpay Payment Integration

- One-time orders for Pro/Max/Family plan purchase
 - Auto-recurring subscriptions (monthly autopay via UPI mandate)
 - B2B seat-based billing for organizations
 - Server-rendered checkout pages for mobile browser
 - HMAC-SHA256 webhook verification (webhooks/razorpay)
- Promo code validation with discount application
 - Refund processing via admin panel
 - Test mode / Live mode toggle via isLiveMode()
 - Family plan with up to 5 member accounts
 - Subscription lifecycle: active != cancelled != expired

Platform Summary

Aorane Health — Complete Technical Overview

100+

API Endpoints

40+

Database Tables

25+

Mobile Screens

26

Admin Pages

13

Business Pages

8

DB Schema Files

Infrastructure at a Glance

- API Server !' Express.js on Render (ap-south-1, Mumbai)
- Admin Panel !' React+Vite on Vercel
- Business Portal !' React+Vite on Vercel
- Landing Website !' React+Vite on Vercel
- Mobile App !' Expo React Native (iOS + Android + Web)
- Database !' Supabase PostgreSQL (Mumbai, 40+ tables)
- Cache !' Upstash Redis (global edge)
- AI !' NVIDIA NIM LLaMA 3.3 70B
- Payments !' Razorpay (orders + subscriptions + webhooks)
- Email !' Resend (transactional)

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